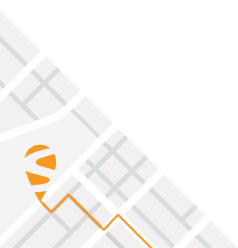




Release Notes and Installation Guide

PolaRx5 Firmware Package v5.7.0



1 Installation Guidelines

To upgrade the firmware to version 5.7.0, upgrade the receiver with the suf file `PolaRx5-5.7.0.suf` as explained in the section **Upgrade the Receiver** of the PolaRx5 Reference Guide.

Upgrading the receiver's failsafe with `PolaRx5-5.7.0-failsafe.suf` is not necessary, unless Septentrio Support advises otherwise.

2 New Features and Improvements

2.1 General Description

To comply with the upcoming EU Radio Equipment Directive (RED) 2014/53/EU, specifically Article 3(3) (d), (e) and (f) cybersecurity requirements, this release contains a set of changes that are introduced to ensure continued regulatory alignment for all our IP-capable receivers. Key changes are highlighted ([security]) in the below sections for visibility.

While maintaining compatibility has been a priority, some modification may impact integration processes, configuration routines, or automated workflows. We recommend reviewing these changes carefully and updating configurations accordingly.

Please note that adjusting default settings is possible but may compromise the effectiveness of the built-in security protections. Users are encouraged to carefully consider the implications of disabling or overriding new defaults, as secure operation remains a shared responsibility.

2.2 New features in version 5.7.0

1. [Security] FTP, SFTP, Rsync and SCP are disabled by default. They can be enabled with a new command `setIPServices`. SSH port is now configurable with `setIPPortSettings`.
2. [Security] In case of lost/forgotten credentials, a mechanism to restore factory default was added.
3. [Security] The Wi-Fi access point now uses WPA2 encryption by default. If not configured, the receiver will generate a password which can be retrieved by authenticated users.
4. [Security] On all web, file transfer and IP interfaces users will by default be required to log in to use and configure the receiver. To support initial log-in, two extra arguments (`FactoryUserName` and `FactoryPassword`) were introduced in the login command.
5. [Security] The Web-interface will automatically redirect to HTTPS, unless it's disabled.
6. [Security] TLS encryption over IP1x connection is now available on port 28783, and can be configured with the `setIPPortSettings` and `setIPServices` commands.
7. The number of supported GPS satellites has been extended to 39 (G01 to G39, with the exclusion of G37).
8. It is now possible to enable observables from the QZSS L1C/B signal in RTCM MSM output messages.

2.3 Improvements in version 5.7.0

1. NTRIP caster authentication with CA chain of trust (i.e. Fingerprint argument of `setNtripTlsSettings` left empty) has been fixed.
2. If Galileo E6C is or has been encrypted on at least one satellite in the last 24 hours, the receiver tracks the E6B signal instead. This applies to all the Galileo satellites even if encryption is only affecting some of them. In the previous release, that time was 10 minutes instead of 24 hours.
3. In GEO Nav file of RINEX version 2.11, incorrect satellite numbering has been fixed
4. The receiver firewall will allow a maximum of 5 ICMP echo-request messages per second on all TCP/IP connections.
5. Rare occurrences of reduced availability of Galileo satellite measurements have been solved.
6. Improved awareness of ionospheric activity when working in static RTK for inter-station distance determination
7. The control of the health flags has been improved. It is now possible to ignore the health flag for specific signals at both measurement and PVT level.

2.4 New features in version 5.6.0

1. Support of dual stack IPv6/IPv4 has been added. As an exception the P2PP feature will remain IPv4 only. The affected commands are: `setIPv6Settings` (new), `setIPServerSettings`, `setIPFiltering`, `setIPReceiveSettings`, `setNMEAFTP`, `setSBFFTP`, `exeFTPPushTest`, `setRINEXFTP`, `setBinexFTP`, `setRTCMMSMFTP`, `exeFTPUpgrade` and `lif`, `ipparameters`. The affected SBF blocks are: `IPv6Status` (new), `WiFiAPIIPv6Status` (new), `WiFiClientStatus` and `DynDNSStatus`.
2. Support for the tracking of the future QZSS L1 C/B signal has been added.
3. Tracking of the new NavIC/IRNSS L1 signal is now supported
4. The number of supported QZSS satellites is extended from 7 to 10. Satellites J08, J09 and J10 are now supported.
5. RINEX v4.01 and v4.02 are now supported (v4.02 includes a new flag field in the GPS CNAV record).
6. Satellites can now be excluded from measurement output based on the OSNMA authentication status.
7. Added option 'RTCMv3' to `setDaisyChainMode` command.
8. The `setRTCMv3Formatting` command has been updated to support setting of maximum total number of satellites in MSM output.
9. Support of the experimental NTS-3 (GPS PRN#34) satellite is available on demand.
10. The receiver now tracks the L5 signal of SBAS satellite S22, even if that signal is not synchronized with the L1 signal.

2.5 Improvements in version 5.6.0

1. The recovery of bit errors in GPS L1C, QZSS L1C and BeiDou B1C nav pages has been improved.

2. A new "IgnoreHealth" option has been added to the setL6CLASSource user command, to support early QZSS L6 experimentations where the Alert/Health flag is raised in the L6 message.
3. A crash of the GNSS receiver, that occasionally could happen when trying to connect to a very congested NTRIP Caster, has been solved.
4. The probability of occurrence of cycle slips has further decreased.
5. An issue affecting the logging of GPS and GLONASS RINEX v2.11 navigation files has been solved.
6. Galileo E5-AltBOC carrier phase is now more robust in non-default configurations.
7. Compatibility with Let's Encrypt TLS certificates has been improved.
8. The sign of the Doppler of the MSS (LBand) geostationary satellites has been aligned with the sign convention of GNSS satellites (positive Doppler when the distance to the satellite decreases).
9. The command exeEchoMessage now also works as expected for IPS and IPR connections.
10. Improved compatibility with OSNMA ICD v1.1
11. Fixed interoperability issue with some NTRIP casters when receiving burst of data.
12. The built-in OSNMA cryptographic keys have been updated to the keys from the OSNMA Public Observation Test Phase that became operational on January 15, 2024.
13. The reporting of satellites below the horizon in the SatVisibility SBF block has been improved, in case a negative elevation mask is configured.
14. The logging is now more robust against manual removal of files when the auto-delete functionality is active.
15. Rare cases of RINEX file corruption after a power outage have been solved.
16. The sensitivity of the spoofing detector has been further improved.
17. Cases of cycle slips when using non-default tracking loop parameters have been solved.
18. BINEX measurement records can now contain data for the QZSS L1C/B signal.
19. In RINEX observation files, the L1C observation type is now replaced by L1W when the GPS L1CA signal is disabled while GPS L1P is enabled.
20. RTCMv3 messages 1032 and 4091 are now correctly counted as accepted in the InputLink SBF block.
21. A rare occurring bug, that sent the receiver on a boot loop, has been fixed.
22. The length of the Server argument of the commands setBinexFtp and setRTCMMSMftp has been increased conform other commands.
23. Removed allowance for non-alphanumeric characters in FTP Upgrade command arguments for stability
24. In the RINEX observation files, the Galileo E6 observation code attribute is set to "X", which needs to be interpreted as "either B or C depending on the E6C encryption status".

3 Known Issues and Limitations

1. If the message "USB device not recognised" is shown on the PC after an upgrade of the receiver firmware, it is recommended to restart the receiver. Afterwards, the USB connection will be functional again.
2. Upgrading from firmware version 5.2.0 (or earlier) to version 5.3.0 (or later), would bring a loss of the SECORX activation, if the receiver has an active SECORX contract. To reactivate the SECORX contract, please contact Septentrio.
3. If more than one user simultaneously changes settings via the web interface, the resulting configuration of the receiver may not be consistent.

4. The time needed to detect an external USB disk connected to the receiver varies with the size of the USB disk and the CPU load of the receiver. For larger USB disks, this can take up to 1 minute.
5. IPS connections may stay visible in the web interface after the client has been disconnected.
6. When connecting to the web interface using clientless SSL VPN, the Expert Control Panel may fail to load commands.
7. The NTRIP server connection is sometimes not reliable when connected to a caster running "Professional Ntrip Broadcaster" (up to v2.0.22).
8. During bootup of the receiver, the status of the external disk is labeled as "no disk detected" in the webserver. Please do not disconnect the disk to avoid filesystem corruption.
9. When logging data at a very low rate, the WebUI Disk Usage widget may not correctly indicate which data types are logged.
10. The Preserve On Event Logging feature only supports preserving of files with a standard IGS filename.
11. Unmounting or remounting the in use disk during logging operations might interrupt post processing actions defined for an active log session.
12. When logging is manually stopped during IGS file runover, the last log file is not added to the compress and FTP push queue.
13. If logging is interrupted due to a power outage and the IGS time frame is passed, the last logging file will not be compressed nor FTP pushed.
14. Static DNS information and DNS information over RA persist even when corresponding network interface is disabled.
15. Identical IPv6 Recursive DNS Server (RDNSS) information received over two different network interfaces may be overwritten.
16. IPv6 Recursive DNS Server (RDNSS) information does not always show up correctly in lif, ipparameters.
17. If the static IPv4 gateway address of the receiver does not fall into the range covered by the netmask, changing the configuration back to DHCP will cause the ethernet interface to fail. A receiver restart is required to restart the ethernet interface.
18. DynDNS.org may delete the IPv6 address when an update is made on a dual-stack network. The IPv4 address will stay valid.
19. When the receiver is used in an IPv6-only network (no concurrent IPv4 connectivity) dynamic DNS resolution does not work.

4 Support

For further information or support, please consult the Septentrio support website (<http://www.septentrio.com/support>).

5 Legal Notice

Septentrio does not authorize the use of its products as critical components in devices or systems intended for safety-of-life applications or in devices or systems, of which the failure may endanger life or cause injuries, unless written approval is given.

All the firmware and documentation delivered with the PolaRx5 Firmware Package is licensed, as explained in the About page which is accessible via the web interface of the receiver. On products without web interface, the License can be viewed with the "IstInternalFile, License" command.

6 System Components and Versions

Product: PolaRx5 Version: 5.7.0 Receiver Platform: SSRC7 Release Date: 04 December 2025	version	PolaRx5-5.7.0-failsafe.suf	PolaRx5-5.7.0.suf
Failsafe	19.0.7-g417c6a1	Y	
Operating System	19.0.7-g417c6a1		Y
GNSS Firmware	2025.08.4-g41a54dd282		Y
Antenna Information	25.2.0		Y